

Solar Tracker using Arduino Uno

Professional Project Report

Abstract

This project presents a prototype dual-axis solar tracker using an Arduino Uno, LDR sensors, and SG90 servo motors. The system continuously aligns a solar panel toward the strongest light source, improving energy capture compared with a fixed panel.

Objectives

- Automatically orient the panel toward sunlight.
- Improve energy collection efficiency.
- Demonstrate embedded control using Arduino and sensors.

Component	Qty	Purpose
Arduino Uno	1	Main controller
LDR	3	Light sensing
SG90 Servo	2	Panel movement
10 kΩ Resistor	3	Voltage divider
Solar Panel	1	Demonstration load

Components Used

Arduino Uno, 3× LDRs, 2× SG90 servo motors, 10 kΩ resistors, jumper wires, breadboard, solar panel, power supply, mounting structure.

Working Principle

LDR sensors detect incident light intensity. The Arduino compares sensor values and generates PWM signals for the servo motors. The servos rotate the panel until the difference between LDR readings is minimized, ensuring the panel faces the brightest direction.

Implementation

The prototype was assembled on a lightweight frame. Voltage divider circuits interface the LDRs to Arduino analog inputs. Servo motors provide horizontal and vertical movement.

Applications

Solar farms, rooftop PV systems, smart street lighting, remote monitoring stations, agricultural automation.

Advantages

Higher energy yield, improved panel efficiency, automated operation, educational value, low-cost implementation.

Limitations

Mechanical wear, higher complexity than fixed systems, power consumption by actuators, reduced benefit on cloudy days.

Future Scope

MPPT integration, ESP32 with IoT monitoring, weather prediction, larger actuators, battery management.

Conclusion

The prototype successfully demonstrates automatic solar tracking using inexpensive electronic components. The project highlights embedded systems, sensor interfacing, and renewable-energy applications while providing a foundation for future improvements.